

HOSPITAL PATIENT FLOW SYSTEM

Software Requirements Specification (SRS)

Document Title	Hospital Patient Flow System — SRS
Version	1.0
Date	March 2026
Status	Draft
Prepared For	Hospital Management
Client	Dr. Dinna

This document describes the functional and non-functional requirements for a digital Hospital Patient Flow Management System.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) v1.0 defines the functional and non-functional requirements for the Hospital Patient Flow Management System deployed as a Multi-Tenant Software as a Service (SaaS) platform. The system serves multiple hospitals simultaneously, where each hospital operates as an independent tenant with complete data isolation from all other hospitals.

1.2 Multi-Tenant SaaS Model

The system operates under the following model:

Layer	Description
System Owner	The company that builds, hosts, and maintains the Hospital Patient Flow System. Manages all hospital tenants through a Super Admin portal.
Hospital Tenant	Each hospital that subscribes to use the system. A hospital is a self-contained tenant with its own data, staff accounts, configurations, and subscription.
Subscription	Each hospital pays a recurring subscription fee (monthly or annual) to access the system. Access is suspended if the subscription lapses.
Data Isolation	Each hospital has its own dedicated database. No data from Hospital A is ever accessible to Hospital B, their staff, or their administrators.

1.3 Deployment Architecture Overview

The recommended architecture uses a separate database per hospital tenant:

- One central application server hosts the system for all tenants.
- Each hospital is provisioned its own isolated database upon subscription activation.
- A tenant identifier routes each login session to the correct hospital database.
- The Super Admin portal connects to a separate master database tracking tenant accounts and subscriptions.
- No queries cross tenant database boundaries under any circumstance.

1.4 Why Separate Database Per Hospital

Factor	Separate DB Per Hospital (Recommended)	Shared DB with Tenant ID
Data isolation	Complete — physically impossible to cross-query	Depends on code — risk of developer error
Patient data safety	Highest — medical data fully separated	Medium — all data in same tables

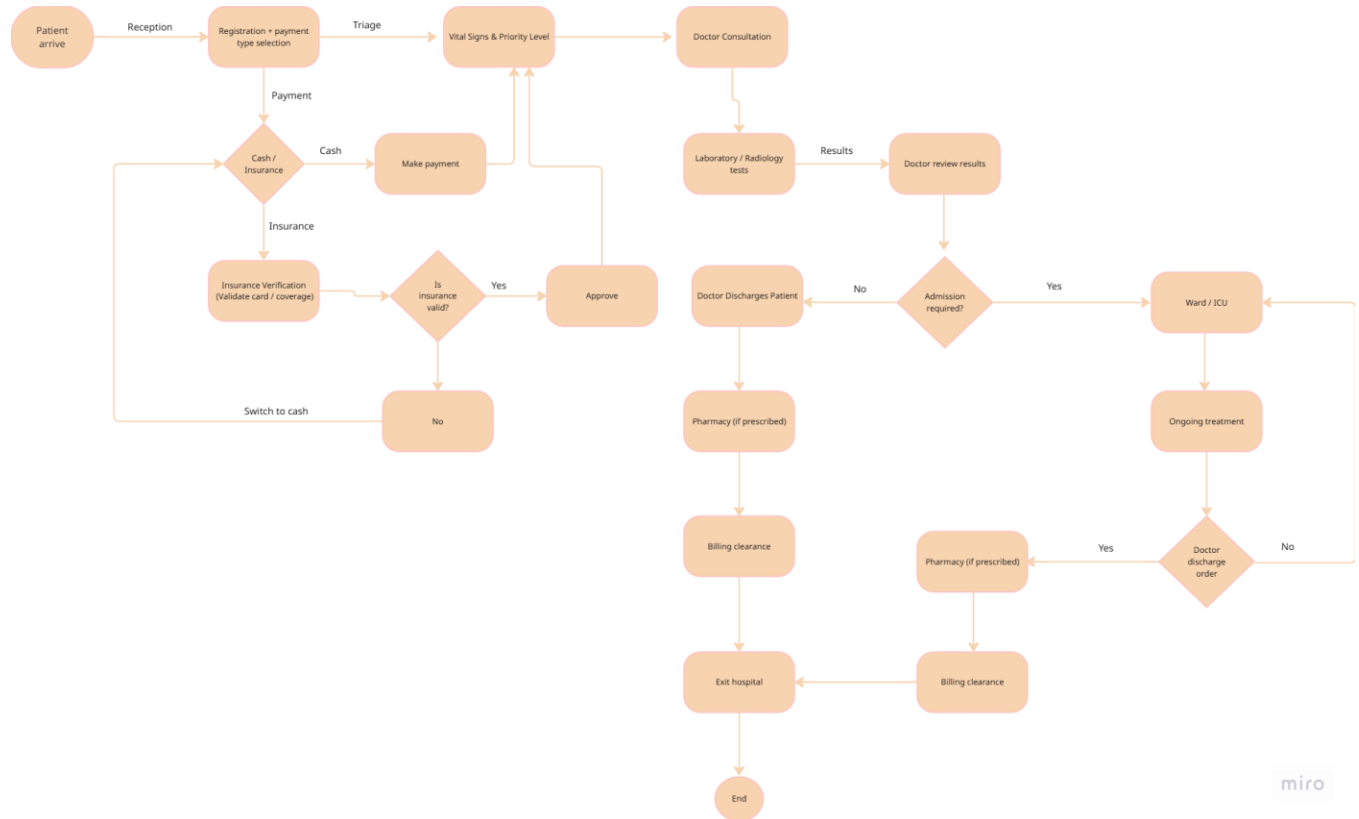
Factor	Separate DB Per Hospital (Recommended)	Shared DB with Tenant ID
Backup & restore	Per hospital — restore one without affecting others	Entire database must be restored
Compliance	Easiest to demonstrate isolation to regulators	Harder to prove isolation
Hospital offboarding	Delete one database cleanly	Complex data extraction needed
Performance	Each hospital's queries do not affect others	Heavy hospitals slow down all tenants
Complexity	More databases to manage	Simpler infrastructure

1.5 System Users

The following user roles interact with the system:

User Role	Scope	Primary Responsibilities
Super Administrator	System-wide (all hospitals)	Onboard new hospitals, manage subscriptions, monitor system health, access no patient data.
Hospital Administrator	Single hospital tenant only	Manage hospital staff accounts, configure departments, fee schedules, and hospital settings.
Receptionist	Single hospital tenant only	Patient registration, payment type selection, queue management.
Triage Nurse	Single hospital tenant only	Vitals recording, triage assessment, priority assignment.
Doctor	Single hospital tenant only	Consultation, diagnosis, investigations, prescriptions, discharge.
Lab Technician	Single hospital tenant only	Test requests, specimen tracking, results entry.
Radiographer	Single hospital tenant only	Imaging requests, report entry, results notification.
Pharmacist	Single hospital tenant only	Prescription dispensing, drug interaction checks, inventory.

1.6 Patient Flow Overview



miro

2. Functional Requirements

Functional requirements define what the system must do. They are organized by module, aligned with each stage of the patient flow.

2.1 Reception Module

2.1.1 Patient Registration

REF	Requirement	Description
FR-01	New Patient Registration	The system shall allow receptionists to register new patients by capturing: full name, date of birth, gender, national ID/passport number, contact details, and next of kin.
FR-02	Returning Patient Lookup	The system shall enable search and retrieval of existing patient records by name, ID number, or patient number.
FR-03	Patient Number Generation	The system shall automatically generate a unique patient number upon registration.
FR-04	Payment Type Selection	The receptionist shall select the patient's payment method: Cash or Insurance (with insurer name and policy number).
FR-05	Insurance Verification	For insurance patients, the system shall validate insurance eligibility and coverage limits in real time or flag for manual verification.
FR-06	Visit Record Creation	The system shall create a new visit/encounter record for each patient attendance, linked to their master record.
FR-07	Queue Assignment	The system shall assign the patient to the triage queue and display their queue number.

2.2 Triage Module

2.2.1 Triage Assessment

REF	Requirement	Description
FR-08	Vital Signs Recording	The nurse shall record patient vital signs: blood pressure, temperature, pulse, oxygen saturation, respiratory rate, and weight.
FR-09	Triage Category Assignment	The system shall support triage classification (e.g. Emergency, Urgent, Semi-Urgent, Non-Urgent) based on recorded vitals and presenting complaint.
FR-10	Chief Complaint Entry	The nurse shall document the patient's presenting complaint in free text and/or from a structured complaint list.
FR-11	Priority Queue Management	The system shall order the doctor's consultation queue based on triage category and time of arrival.
FR-12	Triage Summary	A triage summary shall be available to the attending doctor upon opening the patient's encounter.

2.3 Doctor Consultation Module

REF	Requirement	Description
FR-13	Patient Encounter View	The doctor shall view the full patient record including history, triage summary, vitals, and previous visits.
FR-14	Clinical Notes	The doctor shall document consultation notes including history of presenting illness, examination findings, and clinical impression.
FR-15	Investigation Requests	The doctor shall raise requests for laboratory tests and/or radiology imaging directly within the system.
FR-16	Provisional Diagnosis	The doctor shall record a provisional diagnosis using ICD-10 coding or free text based on the patient's symptoms and clinical examination, before any investigations are requested.
FR-17	Differential Diagnosis	The doctor shall record one or more differential diagnoses using ICD-10 coding or free text, representing alternative conditions being considered alongside the provisional diagnosis.
FR-18	Final / Working Diagnosis	The doctor shall record a final or working diagnosis using ICD-10 coding or free text after reviewing laboratory and/or radiology results. This must be completed before a discharge or admission decision is made.
FR-19	Prescription Issuance	The doctor shall issue prescriptions electronically, specifying drug name, dose, frequency, duration, and route.
FR-20	Referral Management	The doctor shall be able to refer a patient internally (to another department/specialist) or externally.
FR-21	Discharge Decision	The doctor shall mark the patient as Outpatient (for home discharge) or Admission Required.

2.4 Laboratory Module

REF	Requirement	Description
FR-22	Test Request Receipt	Lab technicians shall receive electronic test requests from doctors, showing patient details and tests ordered.
FR-23	Specimen Tracking	The system shall track specimen collection, labelling, and processing status.
FR-24	Results Entry	Lab technicians shall enter test results numerically or categorically, with reference ranges displayed.
FR-25	Critical Value Alerts	The system shall automatically flag and notify the requesting doctor of critical/panic values.
FR-26	Results Availability	Completed results shall be immediately available to the requesting doctor within the patient's encounter.
FR-27	Lab Billing Integration	The system shall generate a charge for each test ordered and route it to the cashier for payment processing.

2.5 Radiology Module

REF	Requirement	Description
FR-28	Imaging Request Receipt	Radiographers shall receive electronic imaging requests specifying modality (X-ray, CT, MRI, Ultrasound) and clinical indication.
FR-29	Imaging Scheduling	The system shall support scheduling of imaging appointments where required.
FR-30	Report Entry	Radiographers or radiologists shall enter findings and impressions as structured or free-text reports.
FR-31	Image Attachment	The system shall support attachment or linking images or scanned reports to the patient's record.
FR-32	Results Notification	The requesting doctor shall be notified when the radiology report is available.
FR-33	Radiology Billing Integration	The system shall generate a charge for each imaging study and route it to the cashier for payment.

2.6 Billing & Cashier Module

REF	Requirement	Description
FR-34	Itemized Bill Generation	The system shall generate an itemised bill for each patient visit covering: consultation fee, investigations, medications, ward charges, and procedures.
FR-35	Multiple Payment Points	The system shall support payment collection at multiple points: before consultation, before lab, before radiology, and at final discharge.
FR-36	Cash Payment Processing	The cashier shall record cash payments and issue a receipt.
FR-37	Insurance Claims Processing	The system shall generate and submit insurance claims electronically, tracking approval/rejection status.
FR-38	Bill Adjustment	Authorized staff shall be able to apply discounts, waivers, or adjustments with a documented reason.
FR-39	Receipt Generation	The system shall print or issue digital receipts for every payment transaction.

2.7 Pharmacy Module

REF	Requirement	Description
FR-40	Prescription Receipt	Pharmacists shall receive electronic prescriptions from doctors, showing drug, dose, frequency, and duration.
FR-41	Drug Dispensing	The pharmacist shall confirm and record drug dispensing against each prescription item.
FR-42	Drug Interaction Check	The system shall alert the pharmacist to known drug-drug interactions or allergy contraindications.
FR-43	Inventory Management	The system shall track drug stock levels and generate low-stock alerts.
FR-44	Dispensing Label Printing	The system shall print dispensing labels with patient name, drug, dose, and instructions.
FR-45	Billing Clearance Verification	The pharmacy module shall verify that billing clearance has been granted before dispensing.
FR-46	Pharmacy Billing	The system shall generate a charge for dispensed medications and link it to the patient's bill.

2.8 Ward / Inpatient Module

REF	Requirement	Description
FR-47	Bed Assignment	The system shall support bed allocation in wards and ICU, displaying bed availability in real time.
FR-48	Admission Record	Upon admission, a formal admission record shall be created with admitting doctor, admitting diagnosis, and ward.
FR-49	Inpatient Orders	Doctors shall issue inpatient orders (medication, nursing, diet, investigations) through the system.
FR-50	Nursing Notes	Nurses shall document observations, interventions, and patient progress notes against the admission record.
FR-51	Discharge Order	The admitting or covering doctor shall issue a formal electronic discharge order with discharge diagnosis and instructions.
FR-52	Length of Stay Tracking	The system shall calculate and display the patient's length of stay for billing and reporting purposes.

2.9 Administration Module

REF	Requirement	Description
FR-53	User Account Management	The administrator shall create, modify, deactivate, and assign roles to system user accounts.
FR-54	Role-Based Access Control	The system shall restrict access to modules and functions based on user role.
FR-55	System Configuration	The administrator shall configure: hospital name, departments, bed counts, fee schedules, and insurance providers.
FR-56	Audit Trail	The system shall maintain a complete, tamper-proof audit log of all user actions with timestamp and user ID.
FR-57	Reports & Analytics	The system shall generate standard reports: patient census, revenue summary, average wait times, bed occupancy, and discharge statistics.
FR-58	Data Backup	The system shall support scheduled automated backups and manual backup initiation by the administrator.

Tenant Context Rule (Applies to All Modules)

TENANT ISOLATION RULE

Every data query, record creation, update, deletion, and report across all modules must be automatically scoped to the logged-in user's hospital tenant. It is architecturally impossible for a user of Hospital A to read, write, or interact with any data belonging to Hospital B. This rule is enforced at the database connection level, not solely at the application code level.

3. Super Administration & SaaS Management

3.1 Super Administration Module

The Super Administrator is an employee of the system owner company. They manage all hospital tenants through a dedicated Super Admin portal that is completely separate from the hospital-facing system.

3.2 Hospital Tenant Management

REF	Requirement	Description
FR-59	Create new hospital tenant	The Super Admin shall onboard a new hospital by entering: hospital name, country, city, address, primary contact name, email, and phone. The system shall automatically provision a new isolated database and assign a unique tenant ID.
FR-60	Configure hospital profile	The Super Admin shall configure each hospital's: time zone, currency, date format, and logo. These settings apply system-wide for that tenant.

REF	Requirement	Description
FR-61	Activate hospital account	The Super Admin shall activate a newly onboarded hospital, triggering welcome email to the hospital administrator and enabling login access.
FR-62	Suspend hospital account	The Super Admin shall suspend a hospital's access (e.g. for non-payment). Suspended hospitals cannot log in. Existing data is retained and not deleted.
FR-63	Reactivate hospital account	The Super Admin shall reactivate a suspended hospital upon payment or resolution of the suspension reason.
FR-64	Terminate hospital account	The Super Admin shall permanently close a hospital account after a defined notice period. Data export must be offered before termination.
FR-65	View all hospital tenants	The Super Admin shall see a dashboard listing all hospitals with their status (Active, Suspended, Trial, Terminated), subscription plan, and subscription expiry date.
FR-66	Impersonate hospital admin	The Super Admin shall be able to view a hospital's system in read-only mode for support purposes. All impersonation sessions must be logged in the audit trail with a clear flag.

3.3 System Monitoring

REF	Requirement	Description
FR-67	System health dashboard	The Super Admin shall view a real-time dashboard showing: system uptime, active users across all tenants, server resource usage (CPU, memory, disk), and any active incidents.
FR-68	Per-tenant usage statistics	The Super Admin shall view usage metrics per hospital: number of active users, patient records created per month, storage consumed, and API call volume.
FR-69	Global audit log	The Super Admin shall access a global audit log of all Super Admin actions including tenant creation, suspension, reactivation, and impersonation.
FR-70	System announcements	The Super Admin shall broadcast system maintenance notices and announcements to all hospital tenants or selected tenants only.

3.4 Subscription & Billing Module

This module manages the commercial relationship between the system owner and each subscribing hospital.

3.5 Subscription Plans

REF	Requirement	Description
FR-71	Subscription plan definition	The system shall support multiple subscription tiers (e.g. Basic, Standard, Premium) each with defined: price, billing cycle (monthly or annual), user seat limit, storage limit, and included modules.
FR-72	Monthly subscription	Hospitals may subscribe on a monthly basis. Payment is due on the same date each month. Access continues month to month until cancelled.
FR-73	Annual subscription	Hospitals may subscribe on an annual basis at a discounted rate. Payment covers 12 months upfront or in agreed instalments.
FR-74	Free trial period	New hospital tenants shall be offered a configurable free trial period (default 30 days) with full system access before a subscription is required.
FR-75	Plan upgrade and downgrade	The Super Admin or hospital administrator shall be able to upgrade or downgrade the subscription plan. Changes take effect at the next billing cycle.
FR-76	Subscription renewal	The system shall automatically generate a renewal invoice before the subscription expiry date (default: 14 days before expiry).

3.6 Payment & Invoicing

REF	Requirement	Description
FR-77	Invoice generation	The system shall automatically generate an itemised invoice for each billing cycle showing: hospital name, plan name, billing period, amount, and due date.
FR-78	Payment recording	The Super Admin shall record payment receipt against each invoice, specifying: payment method, reference number, amount, and date.
FR-79	Payment receipt issuance	Upon payment confirmation, the system shall send a payment receipt to the hospital's registered billing email address.
FR-80	Overdue payment escalation	If payment is not received within 7 days of the due date, the system shall send an automated reminder. If payment is not received within 30 days, the account shall be automatically suspended.
FR-81	Grace period configuration	The Super Admin shall configure the grace period (days after due date before suspension) per tenant if required.
FR-82	Subscription history	The system shall maintain a full history of all invoices, payments, and plan changes for each hospital tenant.

3.7 Hospital Self-Service Subscription Portal

REF	Requirement	Description
FR-83	View subscription status	The hospital administrator shall view their current subscription plan, billing cycle, next payment due date, and account status from within the system.
FR-84	View invoice history	The hospital administrator shall download all past invoices and payment receipts from the subscription portal.
FR-85	Request plan change	The hospital administrator shall submit a plan upgrade or downgrade request to the Super Admin through the portal.
FR-86	Cancellation request	The hospital administrator shall submit a cancellation request with a reason. The Super Admin must confirm before the account is closed.

3.8 Hospital Onboarding Workflow

The following steps define the complete lifecycle from a hospital signing up to becoming fully operational:

Step	Action	Performed By	System Action
1	Hospital enquires and agrees to subscribe	Hospital management	
2	Super Admin creates new tenant account	Super Admin	Provisions isolated database. Assigns tenant ID. Sets trial or subscription plan.
3	Super Admin configures hospital profile	Super Admin	Sets name, country, timezone, currency, logo.
4	Super Admin activates account	Super Admin	Sends welcome email with login credentials to hospital administrator.
5	Hospital Administrator logs in for first time	Hospital Administrator	Forced password change on first login.
6	Hospital Administrator configures departments, wards, fee schedule	Hospital Administrator	System saves configuration to tenant's own database.
7	Hospital Administrator creates staff user accounts	Hospital Administrator	User accounts created, scoped to this tenant only.
8	Staff complete system training	All staff roles	
9	Go-live	All staff	System fully operational for this hospital tenant.
10	Subscription invoice generated at end of trial / billing cycle	System (automated)	Invoice emailed to hospital billing contact.

4. Non-Functional Requirements

Functional requirements define what the system must do. They are organized by module, aligned with each stage of the patient flow.

4.1 Performance

REF	Requirement	Description
NFR-01	Page Load Time	All system pages and screens shall load within 3 seconds under normal operating conditions (up to 100 concurrent users).
NFR-02	Search Response	Patient record searches shall return results within 2 seconds.
NFR-03	Concurrent Users	The system shall support at least 200 concurrent users without performance degradation.
NFR-04	Lab Results Turnaround	Results entered by lab technicians shall be visible to the requesting doctor within 5 seconds.
NFR-05	Report Generation	Standard management reports shall generate within 30 seconds for up to 12 months of data.

4.2 Availability & Reliability

REF	Requirement	Description
NFR-06	System Uptime	The system shall achieve a minimum uptime of 99.5% (excluding scheduled maintenance windows).
NFR-07	Scheduled Maintenance	Maintenance windows shall occur outside peak hours (e.g. 02:00–04:00) and communicated to users 24 hours in advance.
NFR-08	Data Backup Frequency	Automated backups shall run every 6 hours. Full daily backups shall be retained for 30 days.
NFR-09	Disaster Recovery	In the event of system failure, the system shall recover to the last backed-up state within 4 hours (RTO) with no more than 6 hours of data loss (RPO).
NFR-10	Offline Fallback	The system should fallback to local area network (LAN) if available.

4.3 Security

REF	Requirement	Description
NFR-11	Authentication	All users shall authenticate with a unique username and password. Multi-factor authentication (MFA) shall be supported for administrator accounts.
NFR-12	Session Management	User sessions shall automatically time out after 15 minutes of inactivity.
NFR-13	Role-Based Access	No user shall access modules, records, or functions outside their assigned role permissions.
NFR-14	Data Encryption	All patient data shall be encrypted.
NFR-15	Audit Logging	Every data access, creation, modification, and deletion event shall be logged with user ID, timestamp, and IP address.
NFR-16	Password Policy	Passwords shall be a minimum of 8 characters, contain upper and lowercase letters, a number, and a special character. Passwords shall expire every 90 days.
NFR-17	Data Privacy	The system shall comply with applicable personal data privacy regulations, including the <i>Personal Data Protection Act, 2022 (Tanzania)</i> . Personal data shall not be shared with unauthorized third parties.

4.4 Usability

REF	Requirement	Description
NFR-18	Intuitive Interface	The system shall have a role-specific dashboard so each user type sees only the functions relevant to their role upon login.
NFR-19	Training Time	New users in non-technical roles (receptionists, nurses, pharmacists) shall be able to perform core tasks after no more than 4 hours of training.
NFR-20	Error Messages	All error messages shall be written in plain language, clearly describing the problem and suggested corrective action.
NFR-21	Accessibility	The system be accessible 24/7.
NFR-22	Multi-Device Support	The system shall be accessible from desktop computers, laptops, and tablets via a web browser without requiring additional software installation.
NFR-23	Language Support	The system shall support English as the primary language.

4.5 Scalability & Maintainability

REF	Requirement	Description
NFR-24	Scalable Architecture	The system architecture shall support horizontal scaling to accommodate growth in patient volume and additional hospital branches.
NFR-25	Modular Design	The system shall be designed in modules (Reception, Triage, Doctor, Lab, Radiology, Pharmacy, Billing, Ward, Admin) that can be independently updated or extended.
NFR-26	Database Growth	The database shall handle at least 10 years of patient records (estimated 500,000+ records) without performance degradation.
NFR-27	Integration Readiness	The system shall expose APIs to support future integration with external systems such as national health registries, insurance portals, and laboratory instruments.

4.6 Interoperability

REF	Requirement	Description
NFR-28	HL7/FHIR Support	Where applicable, the system shall support HL7 v2.x or FHIR R4 messaging standards for exchange of clinical data.
NFR-29	DICOM Support	The radiology module shall support DICOM standards for imaging data storage and retrieval.
NFR-30	Insurance Portal Integration	The billing module shall support electronic claims submission to major insurance portals via standard API or file format.

4.7 Regulatory Compliance

REF	Requirement	Description
NFR-31	Health Data Regulations	The system shall comply with applicable national health data protection laws and regulations.
NFR-32	Medical Records Retention	Patient records shall be retained for a minimum of 10 years (or as required by local regulation) and shall not be permanently deleted without authorization.
NFR-33	Billing Standards	The billing module shall comply with applicable medical billing codes and standards (e.g. ICD-10 for diagnoses).

4.8 Multi-Tenancy & Data Isolation

REF	Requirement	Description
NFR-34	Complete tenant data isolation	The system shall ensure that no user from Hospital A can ever access, view, modify, or delete data belonging to Hospital B. This isolation shall be enforced at the database level, not solely at the application level.
NFR-35	Separate database per tenant	Each hospital tenant shall have its own dedicated database instance. No patient data, staff records, configuration, or audit logs shall be stored in a shared table with other tenants.
NFR-36	Tenant ID enforcement	Every database connection shall be tied to a specific tenant. The application shall never open a connection that can query across tenant databases simultaneously.
NFR-37	Tenant provisioning time	When a Super Admin creates a new hospital tenant, the system shall complete database provisioning and send the welcome email within 10 minutes.
NFR-38	Tenant suspension enforcement	When a hospital account is suspended, all active sessions for that tenant shall be terminated within 60 seconds. No new logins shall be permitted until reactivation.
NFR-39	Data residency	Where required by regulation, tenant databases shall be hosted in a specified geographic region. The Super Admin shall be able to assign a data region per tenant at onboarding.

4.9 SaaS Availability & Scalability

REF	Requirement	Description
NFR-40	Tenant-independent uptime	A failure or performance issue in one tenant's database shall not cause downtime or degradation for any other tenant.
NFR-41	Horizontal tenant scaling	The system shall support onboarding of new hospital tenants without requiring system downtime or architectural changes.
NFR-42	Maximum tenant capacity	The system shall be designed to support at least 500 hospital tenants simultaneously without performance degradation.
NFR-43	Per-tenant storage limits	Each tenant's storage consumption shall be monitored and enforced according to their subscription plan. Alerts shall be sent to the hospital administrator at 80% and 95% of their storage limit.
NFR-44	Tenant backup independence	Each tenant's database shall be backed up independently. Restoring one hospital's data shall not affect any other hospital's data or operations.

4.10 Subscription Management

REF	Requirement	Description
NFR-45	Automated suspension	The system shall automatically suspend a hospital account exactly 30 days after a payment due date if no payment is recorded. A warning notification shall be sent at 7 days and 3 days before suspension.
NFR-46	Suspension data retention	When a hospital is suspended, all their data must be retained in full for a minimum of 90 days before any deletion is permitted, giving the hospital time to settle payment and reactivate.
NFR-47	Termination data export	Before a hospital account is permanently terminated, the system shall generate and deliver a complete export of all their data in a standard, readable format (CSV/JSON/PDF).
NFR-48	Invoice delivery	Subscription invoices shall be delivered to the hospital's registered billing email within 24 hours of the billing cycle start date.
NFR-49	Subscription audit trail	All subscription events — plan changes, payments, suspensions, reactivations, and terminations — shall be logged with timestamp, actor, and reason.

4.11 Security — Multi-Tenant Additions

REF	Requirement	Description
NFR-50	Super Admin MFA mandatory	Multi-factor authentication (MFA) shall be mandatory for all Super Administrator accounts without exception.
NFR-51	Super Admin action logging	Every Super Admin action — including tenant creation, suspension, reactivation, and impersonation — shall be logged in a tamper-proof global audit trail separate from individual hospital audit logs.
NFR-52	Read-only impersonation	When a Super Admin views a hospital's system for support, they shall have read-only access. They shall not be able to create, modify, or delete any patient or staff data. All impersonation sessions shall display a visible banner indicating the session is being monitored.
NFR-53	Tenant credential isolation	Database credentials for each tenant's database shall be unique per tenant and stored in an encrypted secrets manager. No single credential shall grant access to more than one tenant database.
NFR-54	Penetration testing	The system shall undergo an independent penetration test at least annually, with specific focus on cross-tenant data leakage scenarios. Results and remediation must be documented.

4.12 Compliance — Multi-Tenant Additions

REF	Requirement	Description
NFR-55	Per-tenant data privacy compliance	The system shall be configurable to apply the relevant data retention and privacy rules per tenant.
NFR-56	Right to data portability	Upon request, a hospital tenant shall be able to export all of their data in a portable format within 30 days of the request.
NFR-57	Right to erasure	Upon account termination and after the mandatory retention period, all data for a terminated tenant shall be permanently and irreversibly deleted. A deletion certificate shall be issued.

4.13 Subscription Plan Structure

The following plan structure is proposed for the SaaS subscription offering. Exact pricing is to be defined by management. Plans are designed to accommodate hospitals of different sizes.

Feature	Basic	Standard	Premium
Target hospital size	Small (up to 50 beds)	Medium (50-200 beds)	Large (200+ beds / multi-branch)
User accounts	Up to 20 users	Up to 100 users	Unlimited users
Modules included	Reception, Triage, Doctor, Pharmacy, Basic Billing	All Basic + Lab, Radiology, Full Billing, Reports	All Standard + Ward/ICU, Insurance Integration, Advanced Analytics
Storage per tenant	10 GB	50 GB	200 GB (expandable)
Patient records	Up to 10,000	Up to 100,000	Unlimited
Monthly billing option	Yes	Yes	Yes
Annual billing option	Yes (5% discount)	Yes (10% discount)	Yes (15% discount)
Free trial period	30 days	30 days	30 days
Support level	Email support	Email + Phone support	Dedicated account manager
Uptime SLA	99.0%	99.5%	99.9%
Data backup frequency	Daily	Every 6 hours	Every 1 hour
Custom branding	No	No	Yes (hospital logo on system)

Requirements Summary

5.1 Functional Requirements Summary

Category	v1.0 Count	Ref Range
Reception Module	7	FR-01 – FR-07
Triage Module	5	FR-08 – FR-12
Doctor Consultation	7	FR-13 – FR-19
Laboratory Module	6	FR-20 – FR-25
Radiology Module	6	FR-26 – FR-31
Billing & Cashier	7	FR-32 – FR-38
Pharmacy Module	7	FR-39 – FR-45
Ward / Inpatient	6	FR-46 – FR-51
Administration	6	FR-52 – FR-57
Super Administration	11	FR-58 – FR-68
Subscription & Billing	16	FR-69 – FR-85
Total Functional Requirements	84	FR-01 – FR-85

5.2 Non-Functional Requirements Summary

Category	v1.0 Count	Ref Range
Performance	5	NFR-01 – NFR-05
Availability & Reliability	5	NFR-06 – NFR-10
Security	12	NFR-11 – NFR-17, NFR-50 – NFR-54
Usability	6	NFR-18 – NFR-23
Scalability & Maintainability	4	NFR-24 – NFR-27
Interoperability	3	NFR-28 – NFR-30
Regulatory Compliance	6	NFR-31 – NFR-33, NFR-55 – NFR-57
Multi-Tenancy & Data Isolation	6	NFR-34 – NFR-39
SaaS Availability & Scalability	5	NFR-40 – NFR-44
Subscription Management	5	NFR-45 – NFR-49
Total Non-Functional Requirements	33	NFR-01 – NFR-57

5.3 Overall Totals

Requirement Type	Total
Functional Requirements	85
Non-Functional Requirements	57
Grand Total	142